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ORIGINAL ARTICLE

Surface electromyography and ultrasound evaluation of pelvic floor muscles in hyperandrogenic women

Flávia Ignácio Antonio Vassimon[1] · Cristine Homsi Jorge Ferreira[1] · Wellington Paula Martins[3] · Rui Alberto Ferriani[3] · Roberta Leopoldino de Andrade Batista[1] · Kari Bo[2]

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Abstract
Introduction High levels of androgens increase muscle mass. Due to the characteristics of hyperandrogenism in polycystic ovary syndrome (PCOS), it is plausible that women with PCOS may have increased pelvic floor muscle (PFM) thickness and neuromuscular activity levels compared with controls. The aim of this study was to assess PFM thickness and neuromuscular activity among hyperandrogenic women with PCOS and controls.
Methods This was an observational, cross-sectional, case–control study evaluating PFM by ultrasound (US) and surface electromyography (sEMG) in nonobese women with and without PCOS. Seventy-two women were divided into two groups: PCOS (n=33) and controls (n=39). PFM thickness during contraction was assessed by US (Vingmed CFM 800). Pelvic floor muscle activity was assessed by sEMG (MyoTrac Infinit) during contractions at different time lengths: quick, and 8 and 60 s. Descriptive analysis, analysis of variance (ANOVA), and Student's t test were used for statistical analyses.
Results There were no significant differences in PFM sEMG activity between PCOS and controls in any of the contractions: quick contraction (73.23 mV/ 71.56 mV; p=0.62), 8 s (55.77 mV/ 54.17 mV; p=0.74), and 60 s (49.26 mV/ 47.32 mV; p=0.68), respectively. There was no difference in PFM thickness during contractions evaluated by US between PCOS and controls (12.78 mm/ 13.43 mm; p=0.48).
Conclusions This study did not find statistically significant differences in pelvic floor muscle thickness or in muscle activity between PCOS women and controls.

Keywords Pelvic floor muscles · Polycystic ovary syndrome · Muscle thickness · Muscle activity

Pelvic floor muscle thickness and surface electromyography activity did not differ between women with PCOS and controls

✉ Cristine Homsi Jorge Ferreira
cristine@fmrp.usp.br

[1] Department of Rehabilitation and Functional Performance, Ribeirão Preto Medical School, University of São Paulo, Av. Bandeirantes, 3900 Monte Alegre, 14049-900 Ribeirão Preto, SP, Brazil

[2] Department of Sports Medicine, Norwegian School of Sport Sciences, Oslo, Norway

[3] Department of Obstetrics and Gynecology, Faculty of Medicine of Ribeirão Preto, University of São Paulo, Av. Bandeirantes, 3900 Monte Alegre, 14049-900, Ribeirão Preto, SP, Brazil

Introduction

Androgens and anabolic steroids increase skeletal muscle mass and strength in several muscle groups [1, 2]. The most frequent endocrine abnormality in women during reproductive age is polycystic ovary syndrome (PCOS), with prevalence rates varying from 6 to 21 % [3]. Among other characteristics, hyperandrogenism is present in almost every woman with this syndrome. Hyperandrogenism may lead to an increased muscle size, also including the pelvic floor muscles (PFM) [2]. Due to the characteristics of hyperandrogenism in PCOS, it is plausible that women with PCOS may have increased muscle mass and activity compared with controls.

In a recent study, we found that women with PCOS reported fewer complaints of urinary incontinence (UI) than controls; however, no difference was found in relation to PFM strength measured by vaginal manometry [4]. Several methods can be used to measure different aspects of PFM function, including clinical observation, vaginal palpation, electromyography (EMG), vaginal squeeze-pressure

measurement, urethral pressure measurement, dynamometry, ultrasonography (US), and magnetic resonance imaging (MRI). However, none of the methods are able to evaluate all aspects of PFM function [5].

EMG records bioelectrical activity produced by muscle fibers [6]. Surface EMG (sEMG) has frequently been used to evaluate PFM activation patterns in women with and without pelvic floor dysfunction [7]. US is a noninvasive technique used to evaluate functional and anatomical characteristics of PFM, with accuracy similar to MRI and with proven intra and interobserver reproducibility [8, 9]. This method can provide information on muscle thickness, size, and position. Also, after the exam, images are stored offline and can be analyzed by blind examiners [10]. As there is no gold standard for assessing PFM function and strength, combining more than one method can provide more precise information about PFM function and has been recommended [5]. To date, we have found no report in the literature about PFM evaluation using sEMG and US in women with PCOS. The aim of this secondary analysis was to compare PFM activity and thickness between women with and without PCOS.

Materials and methods

This was an observational, cross-sectional, case–control study evaluating PFM using US and sEMG in women with and without PCOS. The Ethics Committee of the University Hospital, Ribeirão Preto Medical School, University of São Paulo (HC-FMRP-USP) (Protocol HCFMRP n°. 11165/2007) approved the study. All women recruited for the study gave written informed consent to participate.

Participants

From 2008 to 2010, patients scheduled to have a medical consultation at the gynecological endocrinology outpatient or the infertility outpatient clinic were recruited. Women diagnosed with PCOS who fulfilled inclusion criteria were initially identified through their medical records. Nonobese women without PCOS were identified at the infertility outpatient clinic. Weekly, every eligible woman waiting to have a consultation at the clinics was invited to participate. Seventy-nine patients were included. Nonobese [body mass index (BMI)<29.9)] women with PCOS (n=36) were included after confirmation of diagnosis according to the Rotterdam criteria, which require the presence of two of the following: oligo/anovulation, hyperandrogenism clinical (hirsutism or, less commonly, male pattern alopecia), or biochemical [raised free androgen index (FAI) or testosterone >80 ng/dl] [11]. A matched nonobese control group (n=43) was included. Eligibility criterion was being nulliparous with a diagnosis of PCOS. Inclusion criteria for the comparison group were being nulliparous, with no signs or symptoms of hyperandrogenism, and normal ovulatory menstrual cycles (interval 26–33 days). Additional inclusion criteria for both groups were age between 18 and 40 years. Exclusion criteria were a history of prior pregnancy, pelvic surgery, latex allergy, androgen-secreting neoplasms, pituitary adenoma, adrenal hyperplasia, and acromegaly, Cushing syndrome and use of anabolic steroids for autoimmune disease treatment, such as rheumatoid arthritis or systemic lupus erythematosus (SLE). In addition, we excluded women unable to contract the PFM on vaginal palpation using the Oxford Modified Scale and women previously or currently participating in PFM training. Furthermore, women on any medication that might influence muscle mass or strength, including contraceptives and previous treatment for UI, were excluded.

Surface electromyography assessment

Participants were instructed about PFM location and function, with a following assessment using digital vaginal palpation to verify the ability to contract. The active electrode (vaginal probe) was inserted into the vagina and the reference electrode placed in the upper anterior iliac crest. The protocol of assessment consisted of 60 s of rest followed by contractions at different time lengths: five maximum voluntary quick contractions during 2 s (flicks), five contractions held for 8 s, an attempt to hold the contraction for 60 s, and finally 60 s of rest [12]. The equipment used was MyoTrac Infinit with an acquisition frequency of 1 KHz and an accuracy gain of 0.5 %. The intravaginal sensor used was an AS 9572 model, Thought Technology Ltd. ®. The receiver's surface is stainless steel, 27 mm in diameter, and 69 mm in length, with input impedance of 10 $G\Omega^{12}$, frequency band of 10–1 KHz, and a common mode rejection rate (CMRR) >130 dB. The first three flicks were used for subsequent normalization of sEMG data [13, 14]. During signal capture, in each PFM contraction, the abdomen and perineum were observed to determine whether the Valsalva maneuver was performed and if the hip adductor muscles and gluteus muscles were simultaneously contracted. If this occurred, the PFM contraction was disregarded to minimize cross-talk [14]

The sEMG signal was band-pass analog filtered 20–500 Hz and then converted into values of root mean square (RMS). Averages of RMS values were used for data analysis. Mean value of each activity (five flicks, five contraction held for 8 s, and one attempt to hold the contraction for 60 s) were expressed in RMS values and used to obtain one single value [difference between average contraction and average rest (baseline) for each activity]. Single values were then normalized and used for analysis. For sEMG signal normalization, the largest EMG amplitude obtained in one of three maximum voluntary contractions (MVC) was used, as recommended by the Surface ElectroMyoGraphy for the Non-Invasive

Assessment of Muscles project (SENIAM) used in previous studies evaluating PFM [13, 14]. We used three flick PFM MVCs. All means and standard deviations (SD) were normalized, and the difference between mean contractions was calculated.

Ultrasound assessment

All PFM examinations were performed by one physician from the Laboratory of Gynaecology and Obstetrics. Participants were asked to empty the bladder before the exam. To measure PFM thickness, a perineal US situated caudal to the pelvic diaphragm and anterior to the anorectum was applied [15, 16] (Vingmed CFM 800 with a 7.5-MHz vaginal probe). Measurements were standardized as described by Morkved et al. [16], who found an excellent intraexaminer reproducibility [intraclass correlation coefficient (ICC)=0.9841] using this method to evaluate PFM thickness. Three measurements were registered, and values are given as means with 95 % confidence intervals (CI) of the three measurements.

Statistical power and analysis

This study was a secondary analysis, and the power analysis was based only on the primary outcome, which was perineometry [4]. A descriptive analysis was performed to characterize the sample, and analysis of variance (ANOVA) was used to determine the homogeneity of the sample and compare groups regarding continuous variables. Student's *t* test was used to compare means of the two independent groups. All analyses were performed using SAS® 9 software, PROC TTEST. *P* value was set at ≤0.05.

Results

Seventy-nine women were included in the study; however, three from the original PCOS group and four from the control group did not attend EMG and US evaluation appointments and were excluded. Thus, 72 women were divided into the PCOS group [n=33); mean age 27 years (range, 18–36) and mean BMI 23.14 kg/m^2 (range, 16.44–29.9], and the control group [n=39; mean age 30 years (range, 21–40) and mean BMI 23.7 kg/ m^2 (range, 17.3–29.4] for analysis. There were no significant differences in PFM and sEMG in any of the contractions at different time lengths: flicks, and 10 s and 60 s (p=0.62, 0.74, and 0.68, respectively). There was difference in PFM thickness during contraction evaluated using US between groups (Table 1).

Table 1 Comparison of age, body mass index (BMI), and parameters related to surface electromyography (sEMG) and ultrasonography (US) between women with polycystic ovary syndrome (PCOS) and controls

Variables	PCOS (n=33)		Control (n=39)		*P* value
	Mean	SD	Mean	SD	
Age (years)	27	5.5	30	4.9	0.02
BMI (kg/m2)	23.1	3.1	23.7	3.3	0.46
Thickness by US (mm)	12.7	3.6	13.4	3.6	0.48
sEMG flicks (mV)	73.2	13.3	71.5	12.3	0.62
sEMG MVC 8 s (mV)	55.7	21.5	54.1	15.8	0.74
sEMG MVC 60 s (mV)	49.2	20.1	47.3	15.8	0.68

Discussion

The aim of this study was to compare surface EMG (sEMG) of PFM activity and thickness between women with and without polycystic ovary syndrome; no significant difference was found. A search in PubMed found no studies evaluating PFM using sEMG and US in PCOS women. However, since testosterone is an androgen hormone that influences the increase in muscle mass and strength [17], it was suggested in the literature that PCOS women could have increased PFM strength [4]. This hypothesis was not confirmed in the only study found that evaluated PFM strength in PCOS women [4]: Antonio et al. [4] found that the control group had a significantly higher prevalence of UI than the PCOS group, in which no UI was reported, but there was no difference in relation to PFM strength [4]. Mammadov et al. [18] showed that after testosterone administration in female rats with experimentally induced stress urinary incontinence (SUI), there was an increase in leak-point pressure and a larger myofiber cross-sectional area [18]. The level of testosterone in the population studied may be the factor contributing to the absence of UI but not to clinically observable stronger PFM. In humans, a study using MRI showed that increased thickness of the striated urethral sphincter is associated with fewer reports of incontinence episodes and inconvenience in women [19].

Evaluation of PFM function is difficult to perform, and there is no gold standard capable of measuring all functions [20]. In addition, there are no cutoff values for normal muscle function in different age groups. It is recommended that more than one method be used to obtain a comprehensive understanding of PFM function [7, 20]. The literature indicates a moderate correlation between transperineal US and perineometry [7, 21, 22]. A recent study by Pereira et al. [22] using transperineal US found a slight negative correlation between the bilateral diameter of the bulbocavernosus muscle and a maximal perceived voluntary contraction of the PFM using sEMG with a vaginal probe. According to the authors, the moderate to weak correlations found between the different

methods used to evaluate PFM indicates that multiple techniques may be needed for proper assessment [7, 20].

In our study, values in the PCOS and control group obtained through sEMG were 73.23 μV and 71.56 μV, respectively. Comparable values using sEMG reported in the literature in different samples of nulliparous women vary from 20.9 to 90.7 μV [23–25]. Higher values indicate more muscle activity; however, comparing studies is problematic, considering inclusion of different populations and use of different devices and probes [20]. Besides the absence of studies evaluating PFM function in women with PCOS, most studies using EMG to evaluate other populations of women do not describe the use of EMG data normalization [24, 25]. In our study, we performed data normalization as recommended by the SENIAM project. To date, there is no specific consensus in relation to the use of PFM normalization, and important studies in this field have not used this procedure [7, 26]. However, PFM are voluntary contraction muscles, and normalization of EMG data is recommended in order to minimize discrepancies, allowing comparisons between patients, days, muscles, and studies [27, 28]. In spite of this, there are still limitations in the use of sEMG, e.g., the inability to assess a specific muscle activity in isolation, the cross-talk effect, artifact motion, and the moving trend of probes during PFM contractions [20, 29]. In our study, in order to minimize and make these influences homogenous between groups, all procedures during the exams were standardized as much as possible, and only one experienced physiotherapist performed all assessments.

Perineal US allows dynamic representation of examined structures and observation at any angle necessary. This dynamic imaging method plays an important role in the investigation of pelvic floor disorders. We found no difference in PFM thickness measured by US in women with PCOS and controls. This finding corresponds with results of the sEMG reported in our study and of PFM strength reported in our primary analysis [4]. Another study found a strong correlation between PFM strength measured with a manometer and PFM thickness measured by the same US method as the one we used [16].

Our study had one experienced gynecologist performing all exams, and a different study showed excellent intrarater reliability of the US method we used to evaluate PFM thickness [16]. Regardless, results of different US methods cannot be compared, and it is worth noting that we did not use the gold standard US perineal or translabial method to evaluate PFM [30]. The statistical difference of 3 years in age between our two groups may be a confounder. However, the influence of age on PFM is unknown in young women, and we question whether a 3-year difference in nulliparous, untrained women between 27 and 30 years of age is of clinical relevance.

The main limitation of this study is that no power calculations were done for sEMG or US. This study does not provide evidence for substantial difference in PFM activity and thickness between women with PCOS and controls. Future controlled studies with larger sample sizes of multiparous women and using different imaging methods, including the gold standard perineal or translabial US [30], could contribute to the generalization of these findings and elucidate the relationship between hyperandrogenism and different aspects of PFM function.

Conclusion

This study found no significant difference in PFM thickness or sEMG activity between women with PCOS and controls.

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Compliance with ethical standards

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Conflicts of interest None.

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